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LOW HEIGHT CABLE CONNECTOR AND ASSEMBLY THEREOF

Abstract

A cable connector includes a first module, a second module, a third module and a third module. The first module includes a first terminal group and a first cable. The second module includes a second terminal group and a second cable. The third module includes a third terminal group and a third cable. The fourth module includes a fourth terminal group and a fourth cable. The first terminal group and the second terminal group are arranged in a staggered manner along a second direction. The third terminal group and the fourth terminal group are arranged in a staggered manner along the second direction. The first cable and the second cable are located at a first height. The third cable and the fourth cable are located at a 10 second height. The first height is different from the second height. An assembly having the cable connector is also disclosed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] This patent application claims priority of a Chinese Patent Application No. 202410211745.6, filed on Feb. 26, 2024 and titled "CABLE CONNECTOR AND ASSEMBLY THEREOF", the entire content of which is incorporated herein by reference.
TECHNICAL FIELD
[0002] The present disclosure relates to a cable connector and an assembly thereof, which belongs to the technical field of connectors.
BACKGROUND
[0003] A cable connector in the related art usually includes a mounting body and a plurality of modules mounted to the mounting body. The plurality of modules include a first module and a second module. The first module includes a plurality of first conductive terminals and a first cable connected to the plurality of first conductive terminals. The second module includes a plurality of second conductive terminals and a second cable connected to the plurality of second conductive terminals.
[0004] As the requirements for the transmission rate of cable connectors continue to increase, the plurality of modules in the related art need to be expanded, for example, adding a third module and a fourth module. The third module includes a plurality of third conductive terminals and a third cable connected to the plurality of third conductive terminals. The fourth module includes a plurality of fourth conductive terminals and a fourth cable connected to the plurality of fourth conductive terminals. The first cable, the second cable, the third cable and the fourth cable are usually designed in a stacked manner along a height direction of the cable connector.
[0005] As the requirements for the height of the cable connector continue to increase, such stacked cables may easily cause the height of the cable connector to fail to meet the flattening requirements.
[0006] Therefore, it is desirable to improve the cable connector in the related art.
SUMMARY
[0007] An object of the present disclosure is to provide a cable connector and an assembly thereof which can be easily reduced in height.
[0008] In order to achieve the above object, the present disclosure adopts the following technical solution: a cable connector, including: a first module, the first module including a first terminal group and a first cable; the first terminal group including a plurality of first conductive terminals arranged along a first direction; each first conductive terminal including a first contact portion configured to mate with a mating module and a first mounting portion configured to be connected to the first cable; a second module, the second module including a second terminal group and a second cable; the second terminal group including a plurality of second conductive terminals arranged along the first direction; each second conductive terminal including a second contact portion configured to mate with the mating module and a second mounting portion configured to be connected to the second cable; a third module, the third module including a third terminal group and a third cable; the third terminal group including a plurality of third conductive terminals arranged along the first direction; each third conductive terminal including a third contact portion configured to mate with the mating module and a third mounting portion configured to be connected to the third cable; and a fourth module, the fourth module including a fourth terminal group and a fourth cable; the fourth terminal group including a plurality of fourth conductive terminals arranged along the first direction; each fourth conductive terminal including a fourth

contact portion configured to mate with the mating module and a fourth mounting portion configured to be connected to the fourth cable; wherein the first terminal group and the second terminal group are arranged in a staggered manner along a second direction; the third terminal group and the fourth terminal group are arranged in a staggered manner along the second direction; the first cable and the second cable are flush along a third direction and located at a first height; the third cable and the fourth cable are flush along the third direction and located at a second height; the first height is different from the second height; any two of the first direction, the second direction and the third direction are perpendicular to each other.

[0009] In order to achieve the above object, the present disclosure adopts the following technical solution: a cable connector assembly, including: a cable connector including: a first module, the first module including a first terminal group and a first cable; the first terminal group including a plurality of first conductive terminals arranged along a first direction; each first conductive terminal including a first contact portion configured to mate with a mating module and a first mounting portion configured to be connected to the first cable; a second module, the second module including a second terminal group and a second cable; the second terminal group including a plurality of second conductive terminals arranged along the first direction; each second conductive terminal including a second contact portion configured to mate with the mating module and a second mounting portion configured to be connected to the second cable; a third module, the third module including a third terminal group and a third cable; the third terminal group including a plurality of third conductive terminals arranged along the first direction; each third conductive terminal including a third contact portion configured to mate with the mating module and a third mounting portion configured to be connected to the third cable; and a fourth module, the fourth module including a fourth terminal group and a fourth cable; the fourth terminal group including a plurality of fourth conductive terminals arranged along the first direction; each fourth conductive terminal including a fourth contact portion configured to mate with the mating module and a fourth mounting portion configured to be connected to the fourth cable; wherein the first terminal group and the second terminal group are arranged in a staggered manner along a second direction; the third terminal group and the fourth terminal group are arranged in a staggered manner along the second direction; the first cable and the second cable are flush along a third direction and located at a first height; the third cable and the fourth cable are flush along the third direction and located at a second height; the first height is different from the second height; any two of the first direction, the second direction and the third direction are perpendicular to each other; a mating module; the first contact portion, the second contact portion, the third contact portion and the fourth contact portion of the cable connector are all configured to mate with the mating module; and at least one fastener fixing the cable connector and the mating module together.

[0010] Compared with the prior art, the first cable and the second cable of the present disclosure are flush along the third direction and located at a first height. The third cable and the fourth cable are flush along the third direction and located at a second height. The first height is different from the second height. With such arrangement, the first cable, the second cable, the third cable and the fourth cable are prevented from being stacked in sequence, thereby reducing the height of the cable connector and the assembly of the cable connector.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] FIG. **1** is a schematic perspective view of a cable connector assembly in accordance with a first embodiment of the present disclosure;

[0012] FIG. **2** is a perspective view of FIG. **1** from another angle;

[0013] FIG. **3** is a partially exploded perspective view of FIG. **1**;

[0014] FIG. **4** is a partially exploded perspective view of FIG. **3** from another angle;
[0015] FIG. **5** is a partially exploded perspective view of the cable connector of the present disclosure, in which an over-molding housing is separated;
[0016] FIG. **6** is a side view of FIG. **5**;
[0017] FIG. **7** is a further exploded perspective view of FIG. **5**;
[0018] FIG. **8** is a partially exploded view of a first module, a second module, a third module and a fourth module in FIG. **7**, wherein the first module and the second module form one group, and the third module and the fourth module form another group;
[0019] FIG. **9** is a partial enlarged view of a circled portion B in FIG. **8**;
[0020] FIG. **10** is a partial enlarged view of a circled portion C in FIG. **8**;
[0021] FIG. **11** is a further exploded perspective view of FIG. **8**;
[0022] FIG. **12** is a further exploded perspective view of FIG. **11**;
[0023] FIG. **13** is a partial enlarged view of a frame portion D in FIG. **12**;
[0024] FIG. **14** is a partial enlarged view of a frame portion E in FIG. **12**;
[0025] FIG. **15** is a side view of the first module, the second module, the third module and the fourth module in FIG. **7**;
[0026] FIG. **16** is a partial enlarged view of a frame portion F in FIG. **15**;
[0027] FIG. **17** is a partially exploded perspective view of the cable connector of the present disclosure from another angle;
[0028] FIG. **18** is a further exploded perspective view of FIG. **17**;
[0029] FIG. **19** is a perspective view of FIG. **8** from another angle;
[0030] FIG. **20** is a further exploded perspective view of FIG. **19**;
[0031] FIG. **21** is a further exploded perspective view of FIG. **20**; and
[0032] FIG. **22** is a top view of the cable connector assembly in accordance with a second embodiment of the present disclosure.

DETAILED DESCRIPTION

[0033] Exemplary embodiments will be described in detail here, examples of which are shown in drawings. When referring to the drawings below, unless otherwise indicated, same numerals in different drawings represent the same or similar elements. The examples described in the following exemplary embodiments do not represent all embodiments consistent with this application. Rather, they are merely examples of devices and methods consistent with some aspects of the application as detailed in the appended claims.
[0034] The terminology used in this application is only for the purpose of describing particular embodiments, and is not intended to limit this application. The singular forms “a”, “said”, and “the” used in this application and the appended claims are also intended to include plural forms unless the context clearly indicates other meanings.
[0035] It should be understood that the terms “first”, “second” and similar words used in the specification and claims of this application do not represent any order, quantity or importance, but are only used to distinguish different components. Similarly, “an” or “a” and other similar words do not mean a quantity limit, but mean that there is at least one; “multiple” or “a plurality of” means two or more than two. Unless otherwise noted, “front”, “rear”, “lower” and/or “upper” and similar words are for ease of description only and are not limited to one location or one spatial orientation. Similar words such as “include” or “comprise” mean that elements or objects appear before “include” or “comprise” cover elements or objects listed after “include” or “comprise” and their equivalents, and do not exclude other elements or objects. The term “a plurality of” mentioned in the present disclosure includes two or more.
[0036] Hereinafter, some embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. In the case of no conflict, the following embodiments and features in the embodiments can be combined with each other.
[0037] Referring to FIG. **1** to FIG. **21**, the present disclosure discloses a cable connector assembly

which includes a cable connector **100**, a mating module **200** for mating with the cable connector **100**, and at least one fastener **300** for fixing the cable connector **100** and the mating module **200** together.

[0038] Referring to FIG. **3** and FIG. **4**, in an illustrated embodiment of the present disclosure, the mating module **200** is a circuit board. The circuit board includes a first surface **201** (for example, an upper surface), a second surface **202** (for example, a lower surface) opposite to the first surface **201**, and a plurality of conductive holes **203** extending through the first surface **201** and the second surface **202**. It is understandable to those skilled in the art that each conductive hole **203** is a hole in which a metal conductive layer is provided on an inner wall of the hole. The plurality of conductive holes **203** can be connected through traces inside the circuit board.

[0039] Referring to FIG. **3**, in the illustrated embodiment of the present disclosure, the plurality of conductive holes **203** include a plurality of first conductive holes **2031**, a plurality of second conductive holes **2032**, a plurality of third conductive holes **2033** and a plurality of fourth conductive holes **2034**. The plurality of first conductive holes **2031** are arranged in a first row. The plurality of second conductive holes **2032** are arranged in a second row. The plurality of third conductive holes **2033** are arranged in a third row. The plurality of fourth conductive holes **2034** are arranged in a fourth row. The plurality of first conductive holes **2031**, the plurality of second conductive holes **2032**, the plurality of third conductive holes **2033**, and the plurality of fourth conductive holes **2034** are arranged in a staggered manner. That is, the plurality of first conductive holes **2031**, the plurality of second conductive holes **2032**, the plurality of third conductive holes **2033**, and the plurality of fourth conductive holes **2034** are not completely aligned as a whole.

[0040] Specifically, in the illustrated embodiment of the present disclosure, the number of the first conductive holes **2031**, the number of the second conductive holes **2032**, the number of the third conductive holes **2033** and the number of the four conductive holes **2034** are the same. The plurality of first conductive holes **2031** and the plurality of second conductive holes **2032** are arranged in a staggered manner. The plurality of third conductive holes **2033** and the plurality of fourth conductive holes **2034** are arranged in a staggered manner. The plurality of first conductive holes **2031** and the plurality of third conductive holes **2033** are arranged in alignment. The plurality of second conductive holes **2032** and the plurality of fourth conductive holes **2034** are arranged in alignment.

[0041] The circuit board further includes a plurality of first mounting holes **204** which extend through the first surface **201** and the second surface **202**, and are located at four corners of the circuit board. The first mounting hole **204** allows a corresponding fastener **300** to pass through.

[0042] Besides, the circuit board further includes a plurality of positioning holes **205** extending through the first surface **201** and the second surface **202**. The positioning holes **205** are configured to mate with the cable connector **100** to achieve installation positioning.

[0043] Referring to FIG. **4** to FIG. **7**, in the illustrated embodiment of the present disclosure, the cable connector **100** includes a mounting body **5**, a plurality of modules assembled to the mounting body **5**, a conductive pad **6** in contact with a bottom of the mounting body **5**, and an over-molding housing **7** over-molded on the plurality of modules and the mounting body **5**.

[0044] Referring to FIG. **7**, in the illustrated embodiment of the present disclosure, the plurality of modules include a first module **10**, a second module **20**, a third module **30** and a fourth module **40**.

[0045] Referring to FIG. **7** to FIG. **12**, the first module **10** includes a plurality of first terminal groups **11**, a first insulating block **12** fixed on each first terminal group **11**, and a plurality of first cables **13**. Each first terminal group **11** includes a plurality of first conductive terminals **111** arranged along a first direction A**1**-A**1** (for example, a left-right direction). In one embodiment of the present disclosure, the plurality of first conductive terminals **111** are insert-molded with the first insulating block **12**. The first insulating block **12** includes a first top surface **121** and a first bottom surface **122** opposite to the first top surface **121**.

[0046] Each first conductive terminal **111** includes a first contact portion **111***a* configured to mate with the mating module **200** and a first mounting portion **111***b* configured to be connected to the first cable **13**. In the illustrated embodiment of the present disclosure, the first contact portion **111***a* protrudes downwardly beyond the first bottom surface **122**. The first mounting portion **111***b* is upwardly exposed to the first top surface **121**.

[0047] Specifically, as shown in FIG. **9**, in the illustrated embodiment of the present disclosure, the first conductive terminals **111** of each first terminal group **11** include a first ground terminal G**1**, a first signal terminal S**1**, a second signal terminal S**2** and a second ground terminal G**2** which are disposed in sequence along the first direction A**1**-A**1**. Preferably, the first signal terminal S**1** and the second signal terminal S**2** form a differential pair. Each differential pair is associated with two ground terminals on both sides to increase the signal transmission rate and improve the quality of signal transmission.

[0048] Each first cable **13** includes a first ground core wire **131**, a first signal core wire **132**, a second signal core wire **133** and a second ground core wire **134**.

[0049] Specifically, as shown in FIG. **13**, in the illustrated embodiment of the present disclosure, the first mounting portion **111***b* of the first ground terminal G**1** includes a third grounding core wire **131** configured to be electrically connected to the first ground core wire **131**. The first mounting portion **111***b* of the first signal terminal S**1** includes a second mounting surface **111***b***2** configured to be electrically connected to the first signal core wire **132**. The first mounting portion **111***b* of the second signal terminal S**2** includes a third mounting surface **111***b***3** configured to be electrically connected to the second signal core wire **133**. The first mounting portion **111***b* of the second ground terminal G**2** includes a fourth mounting surface **111***b***4** configured to be electrically connected to the second ground core wire **134**.

[0050] The first mounting surface **111***b***1** of the first ground terminal G**1** and the first ground core wire **131** are fixed by soldering or welding. The second mounting surface **111***b***2** of the first signal terminal S**1** and the first signal core wire **132** are fixed by soldering or welding. The third mounting surface **111***b***3** of the second signal terminal S**2** and the second signal core wire **133** are fixed by soldering or welding. The fourth mounting surface **111***b***4** of the second ground terminal G**2** and the second ground core wire **134** are fixed by soldering or welding.

[0051] Referring to FIG. **13**, FIG. **15** and FIG. **16**, in the illustrated embodiment of the present disclosure, the first mounting surface **111***b***1**, the second mounting surface **111***b***2**, the third mounting surface **111***b***3** and the four mounting surface **111***b***4** are located at a same height and located in a first plane M**1**.

[0052] As shown in FIG. **7** to FIG. **12**, the second module **20** includes a plurality of second terminal groups **21**, a second insulating block **22** fixed on each second terminal group **21**, and a plurality of second cables **23**. Each second terminal group **21** includes a plurality of second conductive terminals **211** arranged along the first direction A**1**-A**1**. In one embodiment of the present disclosure, the plurality of second conductive terminals **211** are insert-molded with the second insulating block **22**. The second insulating block **22** includes a second top surface **221** and a second bottom surface **222** opposite to the second top surface **221**.

[0053] Each second conductive terminal **211** includes a second contact portion **211***a* configured to mate with the mating module **200** and a second mounting portion **211***b* configured to be connected to the second cable **23**. In the illustrated embodiment of the present disclosure, the second contact portion **211***a* protrudes downwardly beyond the second bottom surface **222**. The second mounting portion **211***b* is upwardly exposed to the second top surface **221**.

[0054] Specifically, referring to FIG. **9**, in the illustrated embodiment of the present disclosure, the plurality of second conductive terminals **211** of each second terminal group **21** include a third ground terminal G**3**, a third signal terminal S**3**, a fourth signal terminal S**4** and a fourth ground terminal G**4** which are disposed in sequence along the first direction A**1**-A**1**. Preferably, the third signal terminal S**3** and the fourth signal terminal S**4** form a differential pair. Each differential pair is associated with two ground terminals on both sides to increase the signal transmission rate and

improve the quality of signal transmission.
[0055] Each second cable **23** includes a third ground core wire **231**, a third signal core wire **232**, a fourth signal core wire **233** and a fourth ground core wire **234**.
[0056] Specifically, referring to FIG. **13**, in the illustrated embodiment of the present disclosure, the second mounting portion **211***b* of the third ground terminal G**3** includes a fifth mounting surface **211***b***1** configured to be electrically connected to the third ground core wire **231**. The second mounting portion **211***b* of the third signal terminal S**3** includes a sixth mounting surface **211***b***2** configured to be electrically connected to the third signal core wire **232**. The second mounting portion **211***b* of the fourth signal terminal S**4** includes a seventh mounting surface **211***b***3** configured to be electrically connected to the fourth signal core wire **233**. The second mounting portion **211***b* of the fourth ground terminal G**4** includes an eighth mounting surface **211***b***4** configured to be electrically connected to the fourth ground core wire **234**.
[0057] The fifth mounting surface **211***b***1** of the third ground terminal G**3** and the third ground core wire **231** are fixed by soldering or welding. The sixth mounting surface **211***b***2** of the third signal terminal S**3** and the third signal core wire **232** are fixed by soldering or welding.
[0058] The seventh mounting surface **211***b***3** of the fourth signal terminal S**4** and the fourth signal core wire **233** are fixed by soldering or welding. The eighth mounting surface **211***b***4** of the fourth ground terminal G**4** and the fourth ground core wire **234** are fixed by soldering or welding.
[0059] Referring to FIG. **13**, FIG. **15** and FIG. **16**, in the illustrated embodiment of the present disclosure, the fifth mounting surface **211***b***1**, the sixth mounting surface **211***b***2**, the seventh mounting surface **211***b***3** and the eighth mounting surface **211***b***4** are located at a same height and located in the first plane M**1**. In other words, the first mounting surface **111***b***1** of the first ground terminal G**1**, the second mounting surface **111***b***2** of the first signal terminal S**1**, the third mounting surface **111***b***3** of the second signal terminal S**2**, the fourth mounting surface **111***b***4** of the second ground terminal G**2**, the fifth mounting surface **211***b***1** of the third ground terminal G**3**, the sixth mounting surface **211***b***2** of the third signal terminal S**3**, the seventh mounting surface **211***b***3** of the fourth signal terminal S**4**, and the eighth mounting surface **211***b***4** of the fourth ground terminal G**4** are located at the same height and located in the first plane M**1**.
[0060] As shown in FIG. **7** to FIG. **12**, the third module **30** includes a plurality of third terminal groups **31**, a third insulating block **32** fixed on each third terminal group **31**, and a plurality of third cables **33**. Each third terminal group **31** includes a plurality of third conductive terminals **311** arranged along the first direction A**1**-A**1**. In one embodiment of the present disclosure, the plurality of third conductive terminals **311** are insert-molded with the third insulating block **32**. The third insulating block **32** includes a third top surface **321** and a third bottom surface **322** opposite to the third top surface **321**.
[0061] Each third conductive terminal **311** includes a third contact portion **311***a* configured to mate with the mating module **200** and a third mounting portion **311***b* configured to be connected to the third cable **33**. In the illustrated embodiment of the present disclosure, the third contact portion **311***a* protrudes downwardly beyond the third bottom surface **322**. The third mounting portion **311***b* is upwardly exposed to the third top surface **321**.
[0062] Specifically, referring to FIG. **10**, in the illustrated embodiment of the present disclosure, the plurality of third conductive terminals **311** of each third terminal group **31** include a fifth ground terminal G**5**, a fifth signal terminal S**5**, a sixth signal terminal S**6** and a sixth ground terminal G**6** which are disposed in sequence along the first direction A**1**-A**1**. Preferably, the fifth signal terminal S**5** and the sixth signal terminal S**6** form a differential pair. Each differential pair is associated with two ground terminals on both sides to increase the signal transmission rate and improve the quality of signal transmission.
[0063] Each third cable **33** includes a fifth ground core wire **331**, a fifth signal core wire **332**, a sixth signal core wire **333** and a sixth ground core wire **334**.
[0064] Specifically, referring to FIG. **14**, in the illustrated embodiment of the present disclosure,

the third mounting portion **311***b* of the fifth ground terminal G**5** includes a ninth mounting surface **311***b***1** configured to be electrically connected to the fifth ground core wire **331**. The third mounting portion **311***b* of the fifth signal terminal S**5** includes a tenth mounting surface **311***b***2** configured to be electrically connected to the fifth signal core wire **332**. The third mounting portion **311***b* of the sixth signal terminal S**6** includes an eleventh mounting surface **311***b***3** configured to be electrically connected to the sixth signal core wire **333**. The third mounting portion **311***b* of the sixth ground terminal G**6** includes a twelfth mounting surface **311***b***4** configured to be electrically connected to the sixth ground core wire **334**.

[0065] The ninth mounting surface **311***b***1** of the fifth ground terminal G**5** and the fifth ground core wire **331** are fixed by soldering or welding. The tenth mounting surface **311***b***2** of the fifth signal terminal S**5** and the fifth signal core wire **332** are fixed by soldering or welding. The eleventh mounting surface **311***b***3** of the sixth signal terminal S**6** and the sixth signal core wire **333** are fixed by soldering or welding. The twelfth mounting surface **311***b***4** of the sixth ground terminal G**6** and the sixth ground core wire **334** are fixed by soldering or welding.

[0066] Referring to FIG. **14** to FIG. **16**, in the illustrated embodiment of the present disclosure, the ninth mounting surface **311***b***1**, the tenth mounting surface **311***b***2**, the eleventh mounting surface **311***b***3** and the twelfth mounting surface **311***b***4** are located at a same height and located in a second plane M**2**.

[0067] As shown in FIG. **7** to FIG. **12**, the fourth module **40** includes a plurality of fourth terminal groups **41**, a fourth insulating block **42** fixed on each fourth terminal group **41**, and a plurality of fourth cables **43**. Each fourth terminal group **41** includes a plurality of fourth conductive terminals **411** arranged along the first direction A**1**-A**1**. In one embodiment of the present disclosure, the plurality of fourth conductive terminals **411** are insert-molded with the fourth insulating block **42**. The fourth insulating block **42** includes a fourth top surface **421** and a fourth bottom surface **422** opposite to the fourth top surface **421**.

[0068] Each fourth conductive terminal **411** includes a fourth contact portion **411***a* configured to mate with the mating module **200** and a fourth mounting portion **411***b* configured to be connected to the fourth cable **43**. In the illustrated embodiment of the present disclosure, the fourth contact portion **411***a* protrudes downwardly beyond the fourth bottom surface **422**. The fourth mounting portion **411***b* is upwardly exposed to the fourth top surface **421**.

[0069] Specifically, referring to FIG. **10**, in the illustrated embodiment of the present disclosure, the plurality of fourth conductive terminals **411** of each fourth terminal group **41** include a seventh ground terminal G**7**, a seventh signal terminal S**7**, an eighth signal terminal S**8** and an eighth ground terminal G**8** which are disposed in sequence along the first direction A**1**-A**1**. Preferably, the seventh signal terminal S**7** and the eighth signal terminal S**8** form a differential pair. Each differential pair is associated with two ground terminals on both sides to increase the signal transmission rate and improve the quality of signal transmission.

[0070] Each fourth cable **43** includes a seventh ground core wire **431**, a seventh signal core wire **432**, an eighth signal core wire **433** and an eighth ground core wire **434**.

[0071] Specifically, referring to FIG. **14**, in the illustrated embodiment of the present disclosure, the fourth mounting portion **411***b* of the seventh ground terminal G**7** includes a thirteenth mounting surface **411***b***1** configured to be electrically connected to the seventh ground core wire **431**. The fourth mounting portion **411***b* of the seventh signal terminal S**7** includes a fourteenth mounting surface **411***b***2** configured to be electrically connected to the seventh signal core wire **432**. The fourth mounting portion **411***b* of the eighth signal terminal S**8** includes a fifteenth mounting surface **411***b***3** configured to be electrically connected to the eighth signal core wire **433**. The fourth mounting portion **411***b* of the eighth ground terminal G**8** includes a sixteenth mounting surface **411***b***4** configured to be electrically connected to the eighth ground core wire **434**.

[0072] The thirteenth mounting surface **411***b***1** of the seventh ground terminal G**7** and the seventh ground core wire **431** are fixed by soldering or welding. The fourteenth mounting surface **411***b***2** of the seventh signal terminal S**7** and the seventh signal core wire **432** are fixed by soldering or welding. The fifteenth mounting surface **411***b***3** of the eighth signal terminal S**8** and the eighth signal core wire **433** are fixed by soldering or welding. The sixteenth mounting surface **411***b***4** of the eighth ground terminal G**8** and the eighth ground core wire **434** are fixed by soldering or welding.

[0073] In the illustrated embodiment of the present disclosure, by providing the first ground core wire **131** and the second ground core wire **134** on the first cable **13**, by providing the third ground core wire **231** and the fourth ground core wire **234** on the second cable **23**, by providing the fifth ground core wire **331** and the sixth ground core wire **334** on the third cable **33**, and by providing the seventh ground core wire **431** and the eighth ground core wire **434** on the fourth cable **43**, by electrically connecting the ground core wire and the ground terminal and adopting a G-S-S-G arrangement, the cable connector **100** of the present disclosure can meet high-speed applications of 10 GHz.

[0074] Referring to FIG. **14** to FIG. **16**, in the illustrated embodiment of the present disclosure, the thirteenth mounting surface **411***b***1**, the fourteenth mounting surface **411***b***2**, the fifteenth mounting surface **411***b***3** and the sixteenth mounting surface **411***b***4** are located at a same height and located in the second plane M**2**. In other words, the ninth mounting surface **311***b***1** of the fifth ground terminal G**5**, the tenth mounting surface **311***b***2** of the fifth signal terminal S**5**, the eleventh mounting surface **311***b***3** of the sixth signal terminal S**6**, the twelfth mounting surface **311***b***4** of the sixth ground terminal G**6**, the thirteenth mounting surface **411***b***1** of the seventh ground terminal G**7**, the fourteenth mounting surface **411***b***2** of the seventh signal terminal S**7**, the fifteenth mounting surface **411***b***3** of the eighth signal terminal S**8** and the sixteenth mounting surface **411***b***4** of the eighth ground terminal G**8** are located at the same height and located in the second plane M**2**.

[0075] In the illustrated embodiment of the present disclosure, the first terminal group **11** and the second terminal group **21** are arranged in a staggered manner along a second direction A**2**-A**2** (for example, a front-rear direction). The third terminal group **31** and the fourth terminal group **41** are arranged in a staggered manner along the second direction A**2**-A**2**.

[0076] The first cables **13** and the second cables **23** are flush along a third direction A**3**-A**3** (for example, a top-bottom direction) and located at a first height. The third cables **33** and the fourth cables **43** are flush along the third direction A**3**-A**3** and located at a second height. The first height is different from the second height. In the illustrated embodiment of the present disclosure, the first height is higher than the second height.

[0077] In the illustrated embodiment of the present disclosure, any two directions among the first direction A**1**-A**1**, the second direction A**2**-A**2** and the third direction A**3**-A**3** are perpendicular to each other.

[0078] Referring to FIG. **9** and FIG. **10**, in the illustrated embodiment of the present disclosure, the first contact portion **111***a* defines a first through hole **111***a***1**, so that the first contact portion **111***a* has a certain elasticity. The second contact portion **211***a* defines a second through hole **211***a***1**, so that the second contact portion **211***a* has a certain elasticity. The third contact portion **311***a* defines a third through hole **311***a***1**, so that the third contact portion **311***a* has a certain elasticity. The fourth contact portion **411***a* defines a fourth through hole **411***a***1**, so that the fourth contact portion **411***a* has a certain elasticity.

[0079] The first contact portion **111***a* is configured to be inserted into the first conductive hole **2031**, the second contact portion **211***a* is configured to be inserted into the second conductive hole **2032**, the third contact portion **311***a* is configured to be inserted into the third conductive hole **2033**, and the fourth contact portion **411***a* is configured to be inserted into the fourth conductive hole **2034**, so that the cable connector **100** and the mating module **200** are electrically connected.

[0080] Of course, it is understandable to those skilled in the art that in other embodiments of the present disclosure, the first contact portion **111***a*, the second contact portion **211***a*, the third contact portion **311***a* and the fourth contact portion **411***a* can also be designed as elastic arms. At the same time, the first conductive hole **2031**, the second conductive hole **2032**, the third conductive hole **2033** and the fourth conductive hole **2034** can be replaced by metal conductive pads. The cable connector **100** and the mating module **200** can also be electrically connected through the contact between the elastic arms and the metal conductive pads. Since this implementation manner is well known to those skilled in the art, it will not be described in detail in the present disclosure.

[0081] Referring to FIG. **7** and FIG. **18**, in the illustrated embodiment of the present disclosure, the mounting body **5** is a metal shell. The mounting body **5** defines a first receiving groove **51**, a second receiving groove **52**, a third receiving groove **53** and a fourth receiving groove **54** which are disposed in sequence along the second direction A**2**-A**2**. Besides, the mounting body **5** includes a first side wall **511** exposed in the first receiving groove **51**, a second side wall **521** exposed in the second receiving groove **52**, a third side wall **531** exposed in the third receiving groove **53**, and a fourth side wall **541** exposed in the fourth receiving groove **54**. Referring to FIG. **7** and FIG. **18**, in the illustrated embodiment of the present disclosure, an outline of the mounting body **5** is not a regular rectangle. The mounting body **5** includes a first notch **571** on one side and a second notch **572** on the other side. The first notch **571** and the second notch **572** are staggered along the second direction A**2**-A**2**. For example, the first notch **571** is located at a rear of one side, and the second notch **572** is located at a front of the other side. In other words, the mounting body **5** includes a first protrusion **573** on one side and a second protrusion **574** on the other side. The first protrusion **573** and the second protrusion **574** are staggered along the second direction A**2**-A**2**. In the illustrated embodiment of the present disclosure, the first notch **571** and the first protrusion **573** are located on a same side of the mounting body **5**; and the first protrusion **573** is located at a front end of the first notch **571**. The second notch **572** and the second protrusion **574** are located on a same side of the mounting body **5**; and the second protrusion **574** is located at a rear end of the second notch **572**.

[0082] It is understandable to those skilled in the art that the first terminal group **11** and the second terminal group **21** are arranged in the staggered manner along the second direction A**2**-A**2** (for example, the front-rear direction); the third terminal group **31** and the fourth terminal group **41** are arranged in the staggered manner along the second direction A**2**-A**2**. The shape design of the mounting body **5** disclosed in the present disclosure can provide it with good structural strength and avoid failure caused by excessively thin partial wall thickness.

[0083] As shown in FIG. **22**, in another embodiment of the present disclosure, a plurality of the cable connectors **100** can be installed to the mating module **200** in a certain arrangement. For example, the plurality of the cable connectors **100** may be installed to the mating module **200** in a matrix arrangement. Taking two adjacent cable connectors **100** as an example, the second protrusion **574** of one cable connector **100** at least partially protrudes into the first notch **571** of the other cable connector **100**. At the same time, the first protrusion **573** of the other cable connector **100** at least partially protrudes into the second recess **572** of the one cable connector **100**. This arrangement is relatively compact, thereby saving space occupied by the mating module **200**.

[0084] Besides, the mounting body **5** further includes a plurality of second mounting holes **55** located at its four corners. The second mounting holes **55** are aligned with corresponding first mounting holes **204** respectively. The mounting body **5** further includes a plurality of positioning posts **56** protruding downwardly. The positioning posts **56** are configured to be inserted into the positioning holes **205** of the circuit board to achieve installation positioning.

[0085] In the illustrated embodiment of the present disclosure, the two second mounting holes **55** located on a same side of the mounting body **5** are also staggered along the second direction A**2**-A**2**. This arrangement can arrange the second mounting holes **55** at the four outer corners of the mounting body **5** as much as possible. With such arrangement, on the one hand, it increases the installation coverage area and improves the installation reliability; on the other hand, it can minimize the adverse impact on the internal conductive traces of the circuit board caused by opening the first mounting hole **204** on the corresponding circuit board.

[0086] The first module **10** is at least partially received in the first receiving groove **51**. The second module **20** is at least partially received in the second receiving groove **52**. The third module **30** is at least partially received in the third receiving groove **53**. The fourth module **40** is at least partially received in the fourth receiving groove **54**.

[0087] The first ground terminal G**1**, the second ground terminal G**2**, the third ground terminal G**3**, the fourth ground terminal G**4**, the fifth ground terminal G**5**, the sixth ground terminal G**6**, the seventh ground terminal G**7** and the eighth ground terminal G**8** are all in contact with the metal shell in order to achieve better grounding and shielding effects.

[0088] Referring to FIG. **13**, in the illustrated embodiment of the present disclosure, the first ground terminal G**1** includes a first vertical portion **14**. The second ground terminal G**2** includes a second vertical portion **15**. The first vertical portion **14** and the second vertical portion **15** are both in contact with the first side wall **511**.

[0089] The third ground terminal G**3** includes a third vertical portion **24**. The fourth ground terminal G**4** includes a fourth vertical portion **25**. The third vertical part **24** and the fourth vertical part **25** are both in contact with the second side wall **521**.

[0090] As shown in FIG. **14**, the fifth ground terminal G**5** includes a fifth vertical portion **34**. The sixth ground terminal G**6** includes a sixth vertical portion **35**. The fifth vertical portion **34** and the sixth vertical portion **35** are both in contact with the third side wall **531**.

[0091] The seventh ground terminal G**7** includes a seventh vertical portion **44**. The eighth ground terminal G**8** includes an eighth vertical portion **45**. The seventh vertical portion **44** and the eighth vertical portion **45** are both in contact with the fourth side wall **541**.

[0092] Along the first direction A**1**-A**1**, the plurality of the first terminal groups **11** and the plurality of the second terminal groups **21** are alternately arranged. Along the first direction A**1**-A**1**, the plurality of the first cables **13** and the plurality of the second cables **23** are alternately arranged.

[0093] Similarly, along the first direction A**1**-A**1**, the plurality of the third terminal groups **31** and the plurality of the fourth terminal groups **41** are alternately arranged. Along the first direction A**1**-A**1**, the plurality of the third cables **33** and the plurality of the fourth cables **43** are alternately arranged.

[0094] Referring to FIG. **13** and FIG. **14**, in the illustrated embodiment of the present disclosure, the two first terminal groups **11** adjacent along the first direction A**1**-A**1** share a ground terminal; the two second terminal groups **21** adjacent along the first direction A**1**-A**1** share a ground terminal; the two third terminal groups **31** adjacent along the first direction A**1**-A**1** share a ground terminal; the two fourth terminal groups **41** adjacent along the first direction A**1**-A**1** share a ground terminal. The shared ground terminal of at least one of the second terminal groups **21**, the third terminal groups **31** and the fourth terminal groups **41** defines an escape groove **16** for avoiding a corresponding wire.

[0095] Referring to FIG. **7** and FIG. **17**, in one embodiment of the present disclosure, the conductive pad **6** is made of metal material. The conductive pad **6** includes a main body **61** and a plurality of protruding blocks protruding upwardly from the main body **61**. The protruding blocks include a first protruding block **621**, a second protruding block **622**, a third protruding block **623** and a fourth protruding block **624**. The first protruding block **621** is inserted into the first receiving groove **51** from bottom to top to fix the first module **10**. The second protruding block **622** is inserted into the second receiving groove **52** from bottom to top to fix the second module **20**. The third protruding block **623** is inserted into the third receiving groove **53** from bottom to top to fix the third module **30**. The fourth protruding block **624** is inserted into the fourth receiving groove **54** from bottom to top to fix the fourth module **40**. The main body **61** is provided with a mounting surface **611** that abuts against the mating module **200**. The first contact portion **111***a*, the second contact portion **211***a*, the third contact portion **311***a* and the fourth contact portion **411***a* all protrude downwardly beyond the mounting surface **611**. By manufacturing the conductive pad **6** and the mounting body **5** separately, the mounting surface **611** of the conductive pad **6** can be manufactured separately without being affected by the mounting body **5**, thereby reducing costs.

[0096] The main body **61** further includes a plurality of first slots **61***a*, a plurality of second slots **61***b*, a plurality of third slots **61***c* and a plurality of fourth slots **61***d*. The first slots **61***a* are used to allow the first contact portions **111***a* to pass through, so as to adjust the position of the first contact portions **111***a* and improve the accuracy. The second slots **61***b* are used to allow the second contact portions **211***a* to pass through, so as to adjust the position of the second contact portions **211***a* and improve the accuracy. The third slots **61***c* are used to allow the third contact portions **311***a* to pass through, so as to adjust the position of the third contact portions **311***a* and improve the accuracy. The fourth slots **61***d* are used to allow the fourth contact portions **411***a* to pass through, so as to adjust the position of the fourth contact portions **411***a* and improve the accuracy. Through the first slots **61***a*, the second slots **61***b*, the third slots **61***c* and the fourth slots **61***d*, the first contact portions **111***a*, the second contact portions **211***a*, the third contact portions **311***a* and the fourth contact portions **411***a* are capable of performing better position limiting, which is beneficial to improve the insertion accuracy of the first contact portions **111***a*, the second contact portions **211***a*, the third contact portions **311***a* and the fourth contact portions **411***a* into the first conductive holes **2031**, the second conductive holes **2032**, the third conductive holes **2033** and the fourth conductive holes **2034**.
[0097] Besides, the main body **61** further includes a plurality of third mounting holes **612** located at its four corners. The third mounting holes **612** are aligned with corresponding second mounting holes **55** and corresponding first mounting holes **204** to mate with the fasteners **300**. The main body **61** further includes a plurality of positioning notches **613** through which the positioning posts **56** pass.
[0098] In the illustrated embodiment of the present disclosure, the fasteners **300** are bolts, and four bolts are provided. Correspondingly, the second mounting holes **55** and the third mounting holes **612** are provided with internal threads to match external threads of the bolts.
[0099] Compared with the prior art, the first cables **13** and the second cables **23** of the present disclosure are flush along the third direction A**3**-A**3** and located at the first height. The third cables **33** and the fourth cables **43** are flush along the third direction A**3**-A**3** and located at the second height. The first height is different from the second height. This arrangement prevents the first cables **13**, the second cables **23**, the third cables **33** and the fourth cables **43** from being stacked in sequence and occupying a large height, thereby reducing the height of the cable connector **100** and the assembly of the cable connector **100**.
[0100] The above embodiments are only used to illustrate the present disclosure and not to limit the technical solutions described in the present disclosure. The understanding of this specification should be based on those skilled in the art. Descriptions of directions, although they have been described in detail in the above-mentioned embodiments of the present disclosure, those skilled in the art should understand that modifications or equivalent substitutions can still be made to the application, and all technical solutions and improvements that do not depart from the spirit and scope of the application should be covered by the claims of the application.

Claims

1. A cable connector, comprising: a first module, the first module comprising a first terminal group and a first cable; the first terminal group comprising a plurality of first conductive terminals arranged along a first direction; each first conductive terminal comprising a first contact portion configured to mate with a mating module and a first mounting portion configured to be connected to the first cable; a second module, the second module comprising a second terminal group and a second cable; the second terminal group comprising a plurality of second conductive terminals arranged along the first direction; each second conductive terminal comprising a second contact portion configured to mate with the mating module and a second mounting portion configured to be connected to the second cable; a third module, the third module comprising a third terminal group

and a third cable; the third terminal group comprising a plurality of third conductive terminals arranged along the first direction; each third conductive terminal comprising a third contact portion configured to mate with the mating module and a third mounting portion configured to be connected to the third cable; and a fourth module, the fourth module comprising a fourth terminal group and a fourth cable; the fourth terminal group comprising a plurality of fourth conductive terminals arranged along the first direction; each fourth conductive terminal comprising a fourth contact portion configured to mate with the mating module and a fourth mounting portion configured to be connected to the fourth cable; wherein the first terminal group and the second terminal group are arranged in a staggered manner along a second direction; the third terminal group and the fourth terminal group are arranged in a staggered manner along the second direction; the first cable and the second cable are flush along a third direction and located at a first height; the third cable and the fourth cable are flush along the third direction and located at a second height; the first height is different from the second height; any two of the first direction, the second direction and the third direction are perpendicular to each other.
2. The cable connector according to claim 1, wherein the plurality of first conductive terminals comprise a first ground terminal, a first signal terminal, a second signal terminal and a second ground terminal which are disposed in sequence along the first direction; the first cable comprises a first ground core wire, a first signal core wire, a second signal core wire and a second ground core wire; the first mounting portion of the first ground terminal comprises a first mounting surface configured to be electrically connected to the first ground core wire; the first mounting portion of the first signal terminal comprises a second mounting surface configured to be electrically connected to the first signal core wire; the first mounting portion of the second signal terminal comprises a third mounting surface configured to be electrically connected to the second signal core wire; the first mounting portion of the second ground terminal comprises a fourth mounting surface configured to be electrically connected to the second ground core wire; wherein the first mounting surface, the second mounting surface, the third mounting surface and the fourth mounting surface are located in the first plane.
3. The cable connector according to claim 2, wherein the plurality of second conductive terminals comprise a third ground terminal, a third signal terminal, a fourth signal terminal and a fourth ground terminal which are disposed in sequence along the first direction; the second cable comprises a third ground core wire, a third signal core wire, a fourth signal core wire and a fourth ground core wire; the second mounting portion of the third ground terminal comprises a fifth mounting surface configured to be electrically connected to the third ground core wire; the second mounting portion of the third signal terminal comprises a sixth mounting surface configured to be electrically connected to the third signal core wire; the second mounting portion of the fourth signal terminal comprises a seventh mounting surface configured to be electrically connected to the fourth signal core wire; the second mounting portion of the fourth ground terminal comprises an eighth mounting surface configured to be electrically connected to the fourth ground core wire; wherein the fifth mounting surface, the sixth mounting surface, the seventh mounting surface and the eighth mounting surface are located in the first plane.
4. The cable connector according to claim 3, wherein the plurality of third conductive terminals comprise a fifth ground terminal, a fifth signal terminal, a sixth signal terminal and a sixth ground terminal which are disposed in sequence along the first direction; the third cable comprises a fifth ground core wire, a fifth signal core wire, a sixth signal core wire, and a sixth ground core wire; the third mounting portion of the fifth ground terminal comprises a ninth mounting surface configured to be electrically connected to the fifth ground core wire; the third mounting portion of the fifth signal terminal comprises a tenth mounting surface configured to be electrically connected to the fifth signal core wire; the third mounting portion of the sixth signal terminal comprises an eleventh mounting surface configured to be electrically connected to the sixth signal core wire; the third mounting portion of the sixth ground terminal comprises a twelfth mounting surface configured to

be electrically connected to the sixth ground core wire; wherein the ninth mounting surface, the tenth mounting surface, the eleventh mounting surface and the twelfth mounting surface are located in the second plane; the second plane and the first plane have different heights along the third direction.

5. The cable connector according to claim 4, wherein the plurality of fourth conductive terminals comprise a seventh ground terminal, a seventh signal terminal, an eighth signal terminal and an eighth ground terminal which are disposed in sequence along the first direction; the fourth cable comprises a seventh ground core wire, a seventh signal core wire, an eighth signal core wire, and an eighth ground core wire; the fourth mounting portion of the seventh ground terminal comprises a thirteenth mounting surface configured to be electrically connected to the seventh ground core wire; the fourth mounting portion of the seventh signal terminal comprises a fourteenth mounting surface configured to be electrically connected to the seventh signal core wire; the fourth mounting portion of the eighth signal terminal comprises a fifteenth mounting surface configured to be electrically connected to the eighth signal core wire; the fourth mounting portion of the eighth ground terminal comprises a sixteenth mounting surface configured to be electrically connected to the eighth ground core wire; wherein the thirteenth mounting surface, the fourteenth mounting surface, the fifteenth mounting surface and the sixteenth mounting surface are located in the second plane; the cable connector is configured to meet a high-speed application of 10 GHz.

6. The cable connector according to claim 1, wherein the first contact portion defines a first through hole; the second contact portion defines a second through hole; the third contact portion defines a third through hole; the fourth contact portion defines a fourth through hole; the mating module is a circuit board which defines a first conductive hole, a second conductive hole, a third conductive hole and a fourth conductive hole; the first contact portion is configured to be inserted into the first conductive hole; the second contact portion is configured to be inserted into the second conductive hole; the third contact portion is configured to be inserted into the third conductive hole; the fourth contact portion is configured to be inserted into the fourth conductive hole.

7. The cable connector according to claim 1, wherein a plurality of first terminal groups and a plurality of first cables are provided; a plurality of second terminal groups and a plurality of second cables are provided; along the first direction, the plurality of the first terminal groups and the plurality of the second terminal groups are arranged alternately; along the first direction, the plurality of the first cables and the plurality of the second cables are arranged alternately.

8. The cable connector according to claim 7, wherein a plurality of third terminal groups and a plurality of third cables are provided; a plurality of fourth terminal groups and a plurality of fourth cables are provided; along the first direction, the plurality of the third terminal groups and the plurality of the fourth terminal groups are arranged alternately; along the first direction, the plurality of the third cables and the plurality of the fourth cables are arranged alternately.

9. The cable connector according to claim 7, wherein the first module further comprises a first insulating block fixed on each first terminal group so as to connect the first module as a whole; the second module further comprises a second insulating block fixed on each second terminal group so as to connect the second module as a whole; the third module further comprises a third insulating block fixed on each third terminal group so as to connect the third module as a whole; the fourth module further comprises a fourth insulating block fixed on each fourth terminal group so as to connect the fourth module as a whole; the first contact portion extends beyond the first insulating block; the second contact portion extends beyond the second insulating block; the third contact portion extends beyond the third insulating block; the fourth contact portion extends beyond the fourth insulating block; two second terminal groups adjacent along the first direction share one ground terminal; two third terminal groups adjacent along the first direction share one ground terminal; two fourth terminal groups adjacent along the first direction share one ground terminal; at least one ground terminal of the second terminal groups, the third terminal groups and the fourth terminal groups defines an escape groove for avoiding a corresponding wire.

10. The cable connector according to claim 5, further comprising a mounting body; wherein the mounting body comprises a first receiving groove, a second receiving groove, a third receiving groove and a fourth receiving groove which are arranged in sequence along the second direction; the first module is at least partially received in the first receiving groove; the second module is at least partially received in the second receiving groove; the third module is at least partially received in the third receiving groove; the fourth module is at least partially received in the fourth receiving groove.

11. The cable connector according to claim 10, wherein the mounting body is a metal shell; the first ground terminal, the second ground terminal, the third ground terminal, the fourth ground terminal, the fifth ground terminal, the sixth ground terminal, the seventh ground terminal and the eighth ground terminal are all in contact with the metal shell.

12. The cable connector according to claim 11, wherein the first ground terminal comprises a first vertical portion; the second ground terminal comprises a second vertical portion; the mounting body comprises a first side wall exposed in the first receiving groove; the first vertical portion and the second vertical portion are both in contact with the first side wall; the third ground terminal comprises a third vertical portion; the fourth ground terminal comprises a fourth vertical portion; the mounting body comprises a second side wall exposed in the second receiving groove; the third vertical portion and the fourth vertical portion are both in contact with the second side wall; the fifth ground terminal comprises a fifth vertical portion; the sixth ground terminal comprises a sixth vertical portion; the mounting body comprises a third side wall exposed in the third receiving groove; the fifth vertical portion and the sixth vertical portion are both in contact with the third side wall; the seventh ground terminal comprises a seventh vertical portion; the eighth ground terminal comprises an eighth vertical portion; the mounting body comprises a fourth side wall exposed in the fourth receiving groove; the seventh vertical portion and the eighth vertical portion are both in contact with the fourth side wall.

13. The cable connector according to claim 11, further comprising a conductive pad in contact with a bottom of the mounting body; wherein the conductive pad comprises a mounting surface which is close to the mating module; the first contact portion, the second contact portion, the third contact portion and the fourth contact portion all protrude downwardly beyond the mounting surface.

14. A cable connector assembly, comprising: a cable connector comprising: a first module, the first module comprising a first terminal group and a first cable; the first terminal group comprising a plurality of first conductive terminals arranged along a first direction; each first conductive terminal comprising a first contact portion configured to mate with a mating module and a first mounting portion configured to be connected to the first cable; a second module, the second module comprising a second terminal group and a second cable; the second terminal group comprising a plurality of second conductive terminals arranged along the first direction; each second conductive terminal comprising a second contact portion configured to mate with the mating module and a second mounting portion configured to be connected to the second cable; a third module, the third module comprising a third terminal group and a third cable; the third terminal group comprising a plurality of third conductive terminals arranged along the first direction; each third conductive terminal comprising a third contact portion configured to mate with the mating module and a third mounting portion configured to be connected to the third cable; and a fourth module, the fourth module comprising a fourth terminal group and a fourth cable; the fourth terminal group comprising a plurality of fourth conductive terminals arranged along the first direction; each fourth conductive terminal comprising a fourth contact portion configured to mate with the mating module and a fourth mounting portion configured to be connected to the fourth cable; wherein the first terminal group and the second terminal group are arranged in a staggered manner along a second direction; the third terminal group and the fourth terminal group are arranged in a staggered manner along the second direction; the first cable and the second cable are flush along a third direction and located at a first height; the third cable and the fourth cable are flush along the third direction and located at a second height; the first height is different from the second height; any two of the first direction, the second direction and the third direction are perpendicular to each other; a mating module; the first contact portion, the second contact portion, the third contact portion and the fourth contact portion of the cable connector are all configured to mate with the mating module; and at least one fastener fixing the cable connector and the mating module together.

15. The cable connector assembly according to claim 14, wherein the at least one fastener is a bolt.

16. The cable connector assembly according to claim 14, wherein the plurality of first conductive terminals comprise a first ground terminal, a first signal terminal, a second signal terminal and a second ground terminal which are disposed in sequence along the first direction; the first cable comprises a first ground core wire, a first signal core wire, a second signal core wire and a second ground core wire; the first mounting portion of the first ground terminal comprises a first mounting surface configured to be electrically connected to the first ground core wire; the first mounting portion of the first signal terminal comprises a second mounting surface configured to be electrically connected to the first signal core wire; the first mounting portion of the second signal terminal comprises a third mounting surface configured to be electrically connected to the second signal core wire; the first mounting portion of the second ground terminal comprises a fourth mounting surface configured to be electrically connected to the second ground core wire; wherein the first mounting surface, the second mounting surface, the third mounting surface and the fourth mounting surface are located in the first plane.

17. The cable connector assembly according to claim 16, wherein the plurality of second conductive terminals comprise a third ground terminal, a third signal terminal, a fourth signal terminal and a fourth ground terminal which are disposed in sequence along the first direction; the second cable comprises a third ground core wire, a third signal core wire, a fourth signal core wire and a fourth ground core wire; the second mounting portion of the third ground terminal comprises a fifth mounting surface configured to be electrically connected to the third ground core wire; the second mounting portion of the third signal terminal comprises a sixth mounting surface configured to be electrically connected to the third signal core wire; the second mounting portion of the fourth signal terminal comprises a seventh mounting surface configured to be electrically connected to the fourth signal core wire; the second mounting portion of the fourth ground terminal comprises an eighth mounting surface configured to be electrically connected to the fourth ground core wire; wherein the fifth mounting surface, the sixth mounting surface, the seventh mounting surface and the eighth mounting surface are located in the first plane.

18. The cable connector assembly according to claim 17, wherein the plurality of third conductive terminals comprise a fifth ground terminal, a fifth signal terminal, a sixth signal terminal and a sixth ground terminal which are disposed in sequence along the first direction; the third cable comprises a fifth ground core wire, a fifth signal core wire, a sixth signal core wire, and a sixth ground core wire; the third mounting portion of the fifth ground terminal comprises a ninth mounting surface configured to be electrically connected to the fifth ground core wire; the third mounting portion of the fifth signal terminal comprises a tenth mounting surface configured to be electrically connected to the fifth signal core wire; the third mounting portion of the sixth signal terminal comprises an eleventh mounting surface configured to be electrically connected to the sixth signal core wire; the third mounting portion of the sixth ground terminal comprises a twelfth mounting surface configured to be electrically connected to the sixth ground core wire; wherein the ninth mounting surface, the tenth mounting surface, the eleventh mounting surface and the twelfth mounting surface are located in the second plane; the second plane and the first plane have different heights along the third direction.

19. The cable connector assembly according to claim 18, wherein the plurality of fourth conductive terminals comprise a seventh ground terminal, a seventh signal terminal, an eighth signal terminal and an eighth ground terminal which are disposed in sequence along the first direction; the fourth cable comprises a seventh ground core wire, a seventh signal core wire, an eighth signal core wire, and an eighth ground core wire; the fourth mounting portion of the seventh ground terminal comprises a thirteenth mounting surface configured to be electrically connected to the seventh ground core wire; the fourth mounting portion of the seventh signal terminal comprises a fourteenth mounting surface configured to be electrically connected to the seventh signal core wire; the fourth mounting portion of the eighth signal terminal comprises a fifteenth mounting surface configured to be electrically connected to the eighth signal core wire; the fourth mounting portion of the eighth ground terminal comprises a sixteenth mounting surface configured to be electrically connected to the eighth ground core wire; wherein the thirteenth mounting surface, the fourteenth mounting surface, the fifteenth mounting surface and the sixteenth mounting surface are located in the second plane; the cable connector is configured to meet a high-speed application of 10 GHz.

20. The cable connector assembly according to claim 19, wherein the cable connector further comprises a mounting body; wherein the mounting body comprises a first receiving groove, a second receiving groove, a third receiving groove and a fourth receiving groove which are arranged in sequence along the second direction; the first module is at least partially received in the first receiving groove; the second module is at least partially received in the second receiving groove; the third module is at least partially received in the third receiving groove; the fourth module is at least partially received in the fourth receiving groove; wherein the mounting body is a metal shell; the first ground terminal, the second ground terminal, the third ground terminal, the fourth ground terminal, the fifth ground terminal, the sixth ground terminal, the seventh ground terminal and the eighth ground terminal are all in contact with the metal shell.